

S40 NOx Analyser Calibration Report

Notes:

Calibration Performed by	Khanh Nguyen	
Date	28-Aug-13	
Time Begin/End	12:00	18:00
Location	Vinh Yen	

Customer	Thanh Thien
Instrument	Serinus 40
ID No.	13-0472
System/Job No.	500926

Calibration Equipment

Calibrator Model	Gascal 1100
ID/Serial No.	13-0464
Zero Air Source Model	8310LC
ID/Serial No.	13-0466

Gas Std. Cylinder Serial No.	GTA060090
Cylinder Concentration (ppm)	59.20
Cylinder Expiry Date	19-07-14
Cylinder Pressure (Bar)	120
Cylinder Calibration	PASS

Displayed Instrument Parameters

Temperatures

Cell Temp	49.9 - 50.1	50.04
Converter Temp	320 - 330	325.3
Chassis Temp	15 - 45	36.9
PMT	12.95 - 13.05	13.0
Manifold Temp	49.9 - 50.1	49.990

Voltages

High Voltage	640 - 670	659.5
Analogue Supply	11.5 - 12.5	12.0
Digital Supply	4.85 - 5.15	5.0
-10 V	-9.5 - 10.5	-9.9

Pressures and Flow

Ambient		752.160
Cell	70 - 200	164.1
Manifold		705.000
Manifold 2		714.34
Flow	0.6 - 0.66	0.900

Pots

HV Adjust	130 - 150	141.0
PGA Gain	1 - 128	16.0
Input Pot	104	104.0
Test Pot	0	0.0

Firmware Version	2.09.0005
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Instrument Units	ppb
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Single Point Calibration

Initial Span Instrument Gain	0.870
Initial NO Offset (ppb)	0
Initial NO2 Offset (ppb)	0

Full Scale (ppb)	500
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Precalibration Check

Calibration Point	Zero flow (sccm)	Span flow (sccm)	Calculated NOx (ppb)	Measured (ppb)			Error (%FS)
				NO	NO2	NOx	
Zero	3970	0	0	0.370	0.730	1.100	0.2%
80% FS	4441	30.38	402	485.640	-1.550	484.200	16.4%

Note: Pre calibration check shall be within 5% of FS value for NOx span

Zero values should be within 2% of Full Scale

PASS/ FAIL	FAIL
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Complete Filter change and perform leak check

Post-calibration Check

80% FS	4441	30.38	402	397.870	2.260	400.190	-0.4%
Zero	3970	0	0	0.320	0.650	0.970	0.2%

Note: Do not fill in shaded cells

Note: Post calibration check shall be within 1% of FS value for both NOx span and all zeros

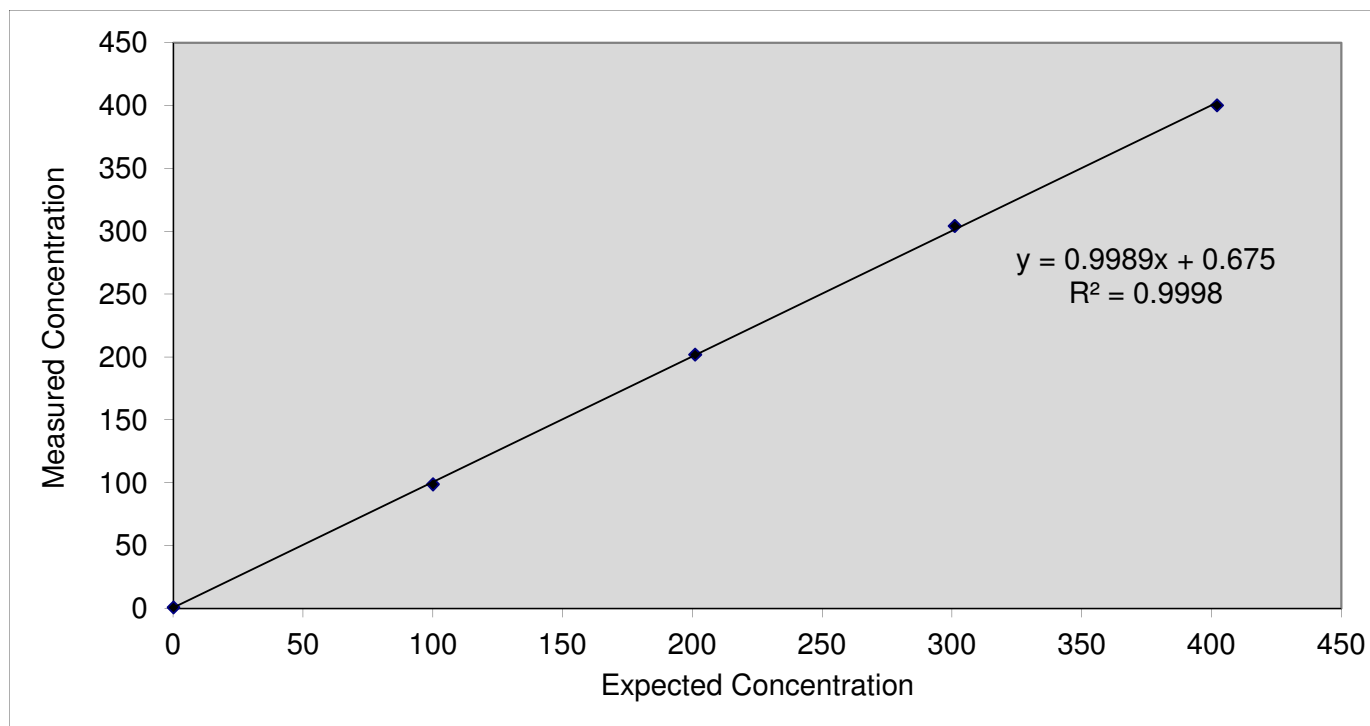
Final Span Instrument Gain	0.960
Final NO Offset (ppb)	0
Final NO2 Offset (ppb)	0

PASS/ FAIL	PASS
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Multipoint Linearity Check

Calibration Point	Zero flow (sccm)	Span flow (sccm)	Calculated NOx (ppb)	Expected NOx (ppb)	Measured NOx (ppb)	Error (%)
Zero	3970	0.00	0	0	0.970	
20% FS	4464	7.58	100	100.000	98.820	-1.2%
40% FS	4457	15.18	201	201.000	202.010	0.5%
60% FS	4451	22.78	301	301.000	304.300	1.1%
80% FS	4441	30.38	402	402.000	400.190	-0.5%
Zero	3970	0.00	0	0	0.530	

Note: Do not fill in the shaded cells



Standard error about $y = x$ (% FS)	0.47%
R-Square Value	0.9998
Intercept	0.6750

PASS/ FAIL **PASS**

Note: Final Zero must be within 1% of FS,
Standard error of y about x must be < 1%FS

Technicians Signature	Date
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